

Lab 3 Report: Clustering, Association Rules, and Recommenders

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Course: DS 7331

Environment: Python 3.11, NumPy, pandas, scikit-learn, seaborn, matplotlib, mlxtend

Data Cards & Ethics Notes

Clustering Dataset: Health & Lifestyle (Kaggle)

- License: CC0 Public Domain
- Shape: 100,000 rows × 16 columns
- Label (used only for evaluation): `disease_risk`

Variable Name	Type	Description	Expected Domain
id	Numeric	Unique identifier for each participant	Positive integers
age	Numeric	Age of the individual (years)	18-90+
gender	Categorical	Gender identity	{Male, Female}
bmi	Numeric	Body Mass Index (kg/m²)	~15-45
daily_steps	Numeric	Number of steps taken per day	≥ 0
sleep_hours	Numeric	Hours of sleep per night	0-12
water_intake_l	Numeric	Daily water intake (liters)	≥ 0
calories_consumed	Numeric	Estimated daily calories consumed (kcal/day)	≥ 0
smoker	Categorical	Smoking status	{0 = No, 1 = Yes}
alcohol	Categorical	Alcohol consumption status	{0 = No, 1 = Yes}
resting_hr	Numeric	Resting heart rate (beats per minute)	≥ 0 (typical adult range ~40-120)
systolic_bp	Numeric	Systolic blood pressure (mmHg)	≥ 0
diastolic_bp	Numeric	Diastolic blood pressure (mmHg)	≥ 0
cholesterol	Numeric	Total cholesterol (mg/dL)	≥ 0
family_history	Categorical	Family history of disease	{0 = No, 1 = Yes}
disease_risk	Target (Categorical)	Overall disease risk classification	{0 = Low Risk, 1 = High Risk}

- Ethical considerations:
 - Fully synthetic; no real health insights should be inferred

- Relationships reflect generator assumptions, not real behaviors
- Safe for algorithm testing but not suitable for medical interpretation

Transactional Dataset: Online Retail (UCI)

- License: CC BY 4.0
- Shape: 541,909 transactions × 8 columns
- ~4,372 customers, ~4,070 items

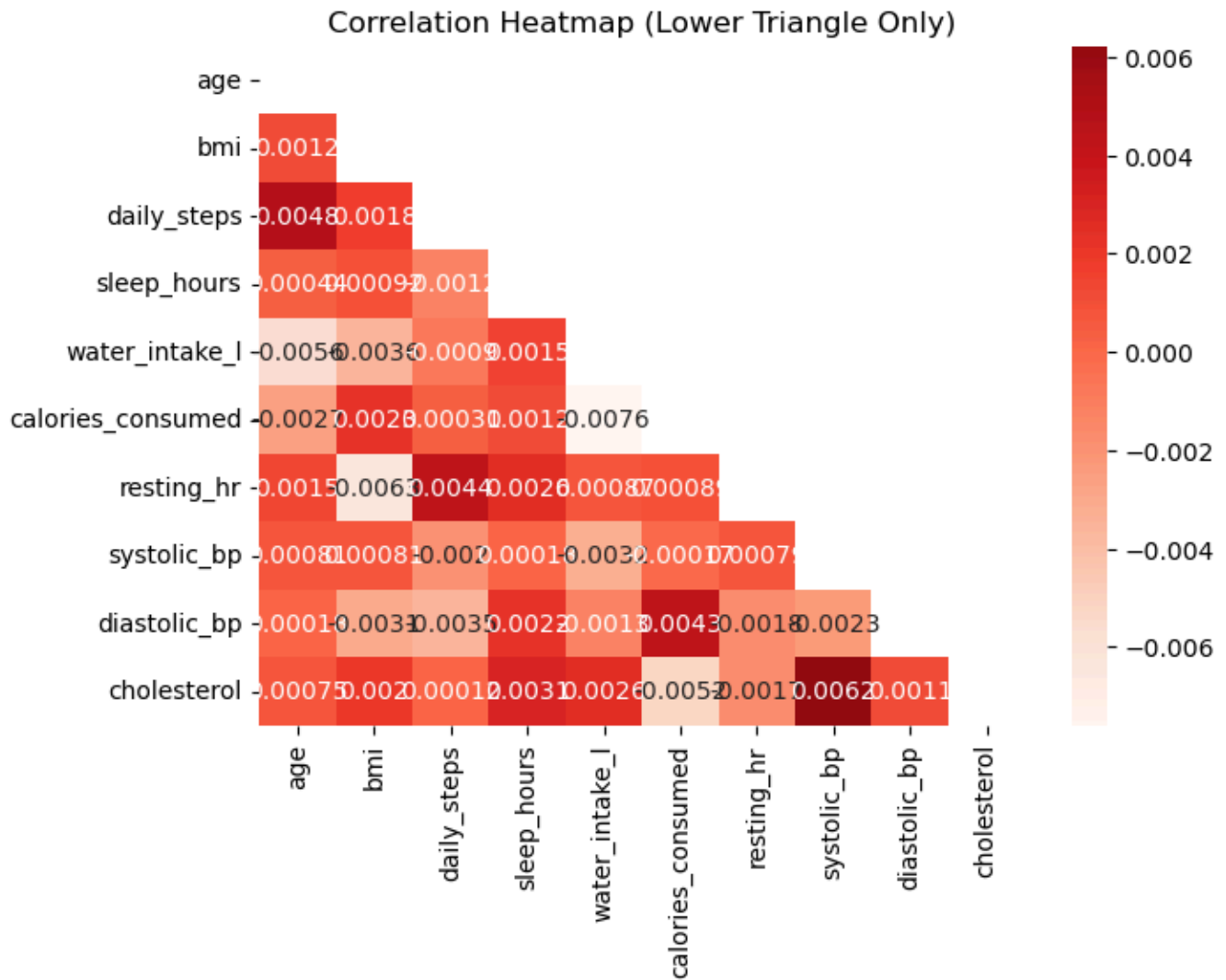
Variable Name	Role	Type	Description
InvoiceNo	ID	Categorical	Invoice number; 6-digit code per transaction. If it starts with letter 'C', the invoice is a cancellation.
StockCode	Item ID	Categorical	Product code uniquely assigned to each distinct product
Description	Feature	Categorical	Product name / description
Quantity	Feature	Integer	Number of units purchased per transaction
InvoiceDate	Feature	Date/Time	Timestamp of transaction
UnitPrice	Feature	Continuous	Product price per unit (£)
CustomerID	User ID	Categorical	5-digit unique customer identifier
Country	Feature	Categorical	Customer's country of residence

- Ethical considerations:
 - Some missing IDs limit user-level completeness
 - Data collected from a single retailer; not generalizable
 - Potential mix of wholesale + retail behaviors

Exploratory Data Analysis

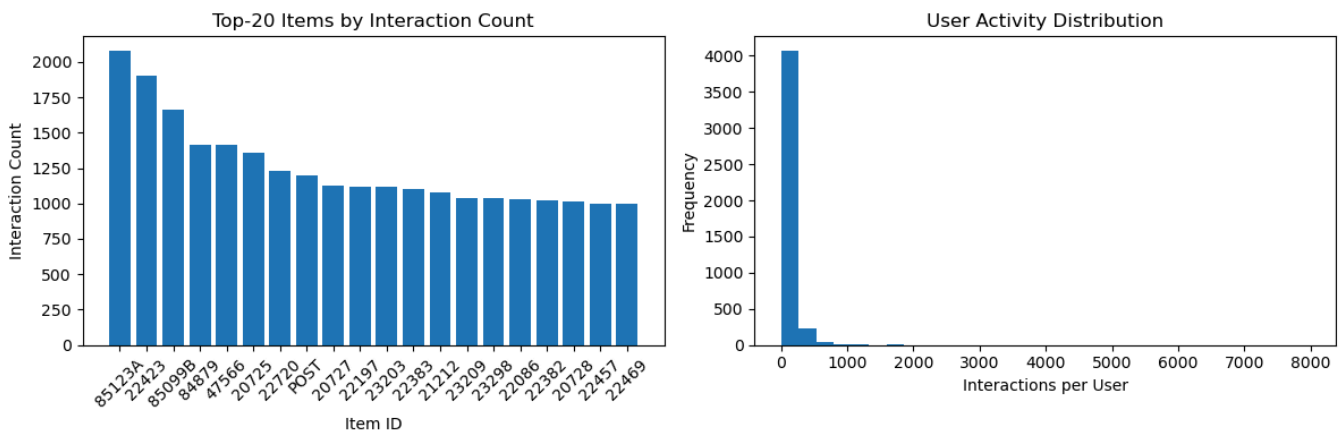
Clustering Dataset

- Numeric features show skewed but smooth distributions.
- Categorical variables are moderately imbalanced but usable.
- Pairwise plots show only slight separation for `disease_risk`.
- **Correlation heatmap:** correlations nearly zero (± 0.006), indicating synthetic independence.
- **Implication:** Very weak natural cluster structure; variables operate on different scales → scaling required.



Transactional Dataset

- **Item popularity:** heavily skewed; few items dominate sales.
- **User activity:** long-tailed; minority of high-activity customers.
- **Co-occurrence:** top-selling items frequently appear in the same carts.
- **Implication:** Strong candidate for association rule mining and item-item recommendations.



Preprocessing Pipeline for Clustering

Raw Data → Drop missing values → OneHotEncoder for categoricals → StandardScaler for numerics
→ Concatenate → PCA for visualization → Compute distance matrices

Clustering Methods and Evaluation

All clustering models were trained on the standardized/scaled, one-hot encoded feature matrix `X_processed`. The `disease_risk` label was never used to train models; it was used only for external evaluation via ARI and NMI. Models were evaluated for $k = 2-8$ (or equivalent parameters) using:

- Internal metrics: Silhouette, Davies–Bouldin, Calinski–Harabasz
- External metrics: Adjusted Rand Index (ARI), Normalized Mutual Information (NMI)
- Visualization: PCA (first two components = 18% variance explained)

Across all methods, internal metrics are low and external metrics remain near zero, indicating weak cluster structure.

K-means

K-means was run for $k = 2-8$.

- Silhouette, Calinski–Harabasz, and Davies–Bouldin consistently favor $k = 2$, although absolute values remain weak.
 - PCA projections show completely overlapping clusters.
 - ARI and NMI remain 0, meaning K-means does not recover `disease_risk` structure.
- Interpretation: If any segmentation exists, it is limited to a very weak 2-cluster partition.

Hierarchical Agglomerative Clustering (HAC)

Two linkages were evaluated: Ward and Average. Dendrograms (300-point subsample) were used for structural interpretation.

- Both linkages show silhouette peaks at $k = 2$ and drift toward zero or negative values as k increases.
 - Dendrograms lack large linkage jumps, suggesting no natural clusters.
 - ARI and NMI remain near zero.
- Interpretation: HAC confirms K-means results — the dataset is weakly clusterable with no clear multicluster structure.

Gaussian Mixture Models (GMM)

GMMs with 2–8 components and full covariance were evaluated.

- Silhouette remains small and best at $k = 2$, confirming overlapping clusters.
- Davies–Bouldin and Calinski–Harabasz show no strong preference for higher k .

- BIC may prefer more components, reflecting density approximation rather than discrete clusters.
 - ARI and NMI remain near zero, so even multiple Gaussian components do not recover disease_risk.
- Interpretation: GMMs support the conclusion that clusters are highly overlapping and weakly separable.

DBSCAN

Explored eps in {0.5, 1.0, 1.5, 2.0, 3.0, 4.0} and min_samples in {3, 5, 10} on a 2,000-point subset.

- Most settings yield one large cluster or mostly noise.
 - A high silhouette (~0.56) at eps = 1.5, min_samples = 3 results from tiny core clusters and nearly all points labeled as noise, making the metric meaningless.
- Interpretation: The dataset lacks density-based structure; DBSCAN is not suitable for this problem.

Spectral Clustering

Used a 10-nearest-neighbor affinity graph and evaluated k = 2–8 clusters.

- Best silhouette at k = 2, but values remain low.
 - External metrics remain near zero.
 - PCA plots show overlapping clouds similar to K-means and GMM.
- Interpretation: Spectral clustering does not uncover meaningful manifold structure.

Graph Clustering (Connected Components)

Constructed a 10-nearest-neighbor graph.

- The graph forms a single connected component.
 - Internal metrics are undefined in this case.
- Interpretation: Graph-based clustering confirms that the data form one continuous connected region.

Clustering Summary

The table below summarizes best-performing hyperparameters and internal/external metrics across methods.

Method	Best Params	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
KMeans	n_clusters=2, n_init=10, random_state=42	0.070	3.567	385.259	-0.000	0.000
HAC (Ward)	n_clusters=2, linkage=ward, metric=euclidean	0.028	4.990	155.883	0.001	0.000

Method	Best Params	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
HAC (Average)	n_clusters=2, linkage=average, metric=euclidean	0.123	1.173	2.964	-0.001	0.000
GMM	n_components=2, covariance_type=full, random_state=42	0.067	3.617	374.655	0.000	0.000
Spectral Clustering	n_clusters=2, assign_labels=kmeans, random_state=42	0.062	3.816	131.355	0.003	0.000
DBSCAN	eps=1.5, min_samples=3	nan	nan	nan	nan	nan
Graph Clustering (kNN components)	n_neighbors=10, affinity=kNN graph	nan	nan	nan	nan	nan

Association Rule Mining

Frequent Itemsets

- After filtering (support $\geq 2\%$): 209 items retained.
- Apriori (2%): 251 itemsets, max length = 3.
- Apriori (3%): 94 itemsets, all singletons.
- FP-Growth returns identical itemsets; runtime slightly slower due to dense matrix.

Association Rules (Support = 0.02, Confidence ≥ 0.30)

- 79 rules generated; average rule length ≈ 2.1 .
- No rules generated at 3% support.

Key Pattern: Strong 3-Item Product Trio

Items **22697**, **22698**, and **22699** exhibit:

- Very high lift (= 24)
- High confidence and high leverage
- Dominance across top-ranked rules
- These suggest a cohesive product family

Potentially Misleading Rules

- High confidence + lift equals 1 $\rightarrow$ reflects popularity, not association.
- Very low support rules $\rightarrow$ unstable and not actionable.

Threshold Sensitivity

- Increasing support (0.02 to 0.03): 251 itemsets $\rightarrow$ 94; rules: 79 $\rightarrow$ 0.

- Higher confidence: fewer, more trivial rules.
- Best interpretability occurs at 2% support.

Conclusions: Key Takeaways, Risks, and Next Steps

Key Takeaways

- The synthetic health and lifestyle dataset does not exhibit strong natural clusters.
 - All clustering methods (K-means, HAC, GMM, DBSCAN, spectral, and graph-based) agree on a very weak 2-cluster structure.
 - Internal metrics are modest at best, and external metrics (ARI, NMI) indicate that clusters do not meaningfully align with the disease_risk label.
 - Disease risk in this dataset appears to vary smoothly rather than forming distinct segments.
- In contrast, the Online Retail dataset contains clear and actionable association patterns.
 - Strong co-purchasing signals emerge, particularly for a specific trio of products (StockCodes 22697, 22698, 22699).
 - These associations are highly actionable for bundling, recommendation systems, and merchandising.
- Threshold tuning is crucial for association mining.
 - A 2% support threshold captures meaningful multi-item patterns, while 3% support collapses all itemsets to singletons and eliminates rule generation.
 - Confidence thresholds control the balance between novelty and interpretability.

Risks

- Synthetic clustering data lacks real-world interpretability.
- Association rules may be biased by item popularity or low support.
- FP-Growth vs. Apriori performance varies with matrix density.

Next Steps

- Explore nonlinear methods such as UMAP or t-SNE.
- Engineer domain-informed features and consider supervised models for prediction.
- Add variables like price, seasonality, and quantity to association analyses.
- Lower support slightly (<2%) to capture additional meaningful rules.
- Use rules to develop an item-item recommender.
- Combine discovered rules with segmentation strategies.
- Validate unusual or high-impact rules with product metadata.